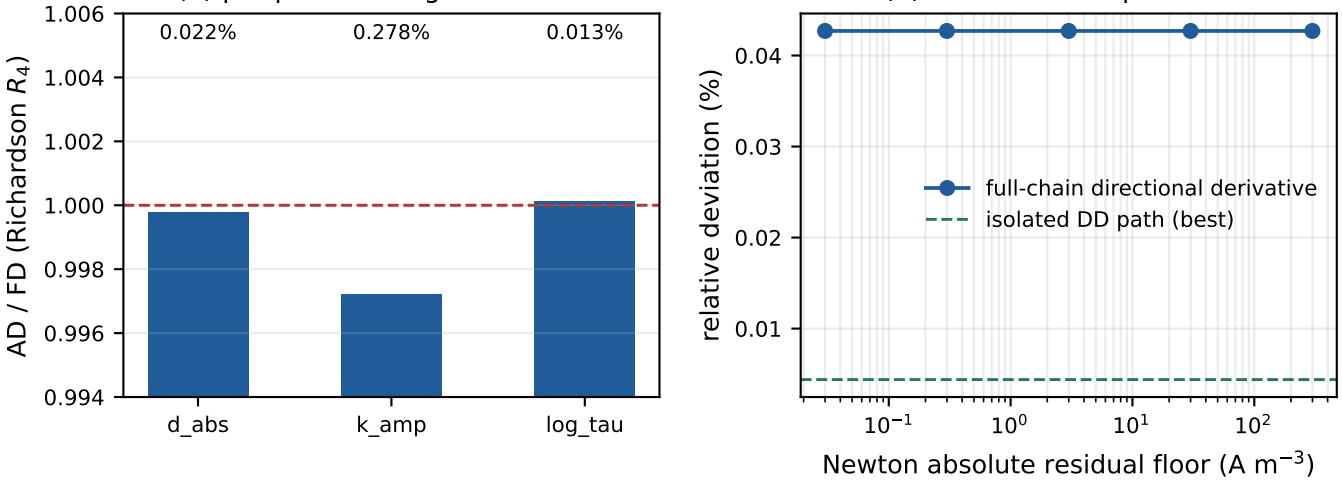
(a) per-parameter gradient ratio
AD / FD (Richardson R_4)
1.006
1.004
1.002
1.000
0.998
0.996
0.994
0.022%
0.278%
0.013%
d_abs
k_amp
log_tau
(b) tolerance independence
relative deviation (%)
0.04
0.03
0.02
0.01
full-chain directional derivative
isolated DD path (best)
10^{-1}
10^{0}
10^{1}
10^{2}
Newton absolute residual floor (A m^{-3})